Homework 5

Module 5 - Single Oral Administration · Homework · No date

Name _____ Date _____

1. A single 500mg dose of medication was administered orally. The medication is about 88% orally bioavailable and is characterized as having an absorption half-life of approximately 75 minutes, elimination half-life of approximately 4 hours, and apparent volume of distribution of 25 L.

1a When does the maximum concentration of drug in the plasma occur?

1b What is the expected maximum concentration following this single oral dose?

1c What is the expected AUC following this single oral dose?

1d If the dose were increased to 1000 mg, when does the maximum concentration of drug in the plasma occur?

1e If the dose were increased to 1000 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f If the dose were increased to 1000 mg, what is the AUC that you would expect?

2. A 78 kg male patient received multiple IV bolus injections of 15 mg/kg every 8 hours for 48 hours. The medication has an elimination half-life of 4 hours and an apparent

volume of distribution that is 22% of body weight.

2a What is the average concentration of drug in the plasma at steady state?

2b What is the maximum concentration of drug in the plasma at steady state?

2c What is the minimum concentration of drug in the plasma at steady state?

2d What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?